

2013-2014 学年第二学期期末考试试卷

1. $x \frac{dy}{dx} + (x+1)y = 3x^2 e^{-x} (x > 0)$

2. $2xy^3 dx = (1 - x^2 y^2) dy$

$$3. (y + xy^2)dx + (x - x^2y)dy = 0$$

$$4. 2y \frac{d^2 y}{dx^2} - \left(\frac{dy}{dx} \right)^2 = 0, y(1) = 1, \left. \frac{dy}{dx} \right|_{x=1} = 2$$

5. $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = e^{2x}$

6. $(1 - x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 2y = 0$

$$7. x^2 \frac{dy}{dx} + 4x \frac{dy}{dx} + 2y = \frac{1}{x+1}$$

8. 函数 $f(x)$ 具有二阶导数且二阶导数连续, 试求满足 $\int_0^x (x-t)f(t)dt = \cos x - f(x)$ 的函数 $f(x)$

$$9. \begin{cases} \frac{dx}{dt} = y + z \\ \frac{dy}{dt} = x + z \\ \frac{dz}{dt} = x + y \end{cases}$$

2013-2014 学年第二学期期末考试试卷参考答案

1. 【解析】方程即为 $\frac{dy}{dx} = -\left(1 + \frac{1}{x}\right)y + 3xe^{-x}$

$$y = e^{\int -\left(1 + \frac{1}{x}\right)dx} \left(\int 3xe^{-x} e^{\int \left(1 + \frac{1}{x}\right)dx} dx + c \right) = \frac{1}{xe^x} \left(\int 3xe^{-x} xe^x dx + c \right)$$

$$= \frac{1}{xe^x} \left(\int 3x^2 dx + c \right) = \frac{1}{xe^x} (x^3 + c) = e^{-x} \left(x^2 + \frac{c}{x} \right)$$

【考点延伸】《考试宝典》专题六 题型 2——一阶微分方程的求解

2. 【解析】方程即为 $2xy^3 dx + (x^2 y^2 - 1) dy = 0$, 令 $M = 2xy^3, N = x^2 y^2$

$$\varphi(y) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = -\frac{2}{y}, \mu = e^{\int \varphi(y) dy} = \frac{1}{y^2}, \text{由积分因子法可得方程化为}$$

$$2xy dx + \left(x^2 - \frac{1}{y^2} \right) dy = 0, \text{由此解得方程: } x^2 y + \frac{1}{y} = c$$

【考点延伸】《考试宝典》专题六 题型 2——一阶微分方程的求解

3. 【解析】法一: $\frac{dy}{dx} = \frac{xy^2 + y}{x^2 y - x} = \frac{y(xy + 1)}{x(xy - 1)}$, 令 $u = xy$, 则 $\frac{dy}{dx} = \frac{d\left(\frac{u}{x}\right)}{dx} = \frac{1}{x} \frac{du}{dx} - \frac{1}{x^2} u = \frac{u+1}{u-1} \frac{u}{x^2}$

所以 $\frac{du}{dx} = \frac{1}{x} \frac{2u^2}{u-1} \Rightarrow \left(\frac{1}{u} - \frac{1}{u^2} \right) du = 2 \frac{dx}{x}$, 两边积分得 $\ln u + \frac{1}{u} = 2 \ln x + c$, 即 $\ln \frac{y}{x} + \frac{1}{xy} = c$,

法二: 令 $u = \frac{y}{x}$, 则 $x \frac{du}{dx} + u = \frac{ux^2 + 1}{ux^2 - 1} u \Rightarrow \frac{du}{dx} = \frac{2u}{ux^3 - x}$, $\frac{dx}{du} = \frac{1}{2} \left(x^3 - \frac{x}{u} \right)$

故 $\frac{2}{x^3} \frac{dx}{du} = 1 - \frac{1}{ux^2}$. 令 $z = \frac{1}{x^2}$, 则 $\frac{dz}{du} = \frac{z}{u} - 1$, 故 $z = e^{\int \frac{1}{u} du} \left(\int -e^{\int -\frac{1}{u} du} du + c \right) = u(-\ln u + c)$

所以 $\frac{1}{x^2} = \frac{y}{x} \left(-\ln \frac{y}{x} + c \right)$, 即 $\frac{1}{xy} + \ln \frac{y}{x} = c$

【考点延伸】《考试宝典》专题六 2.1—可分离变量的微分方程

4. 【解析】 $p = \frac{dy}{dx}$, 则 $\frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dy}$, 故 $2yp \frac{dp}{dy} - p^2 = 0 \Rightarrow \frac{dp}{dy} = \frac{1}{2y} p$, $p = ce^{\int \frac{1}{2y} dy} = c\sqrt{|y|}$

因为 $y(1) = 1$, $\left. \frac{dy}{dx} \right|_{x=1} = 2$, 故 $c = 2$, 因此 $\frac{dy}{dx} = 2\sqrt{|y|} \Rightarrow \frac{dy}{2\sqrt{|y|}} = dx$, $\sqrt{|y|} = x + c_0$

代入 $y(1) = 1$, 得 $c_0 = 0$, 故 $\sqrt{|y|} = x$, $y = x^2$

【考点延伸】《考试宝典》专题六 3.3— $y'' = f(y, y')$ 型微分方程

5. 【解析】特征方程为 $\lambda^2 - 4\lambda + 4 = 0 \Rightarrow \lambda = 2$ (二重根), 故 $Y = c_1 e^{2x} + c_2 x e^{2x}$

令 $y^* = A x^2 e^{2x}$, 代入得 $A = \frac{1}{2}$, 故 $y^* = \frac{1}{2} x^2 e^{2x}$, 因此 $y = Y + y^* = \left(c_1 + c_2 x + \frac{1}{2} x^2 \right) e^{2x}$

【考点延伸】《考试宝典》专题六 4.3—二阶常系数线性微分方程的求解

6. 【解析】可知该方程的特解为 $y_1 = x$, 则由 Liouville 公式可得通解为:

$$\begin{aligned} y &= y_1 \left(c_1 + c_2 \int \frac{1}{y_1^2} e^{\int -\frac{2x}{1-x^2} dx} dx \right) = x(c_1 + c_2) \int \frac{1}{x^2} (1-x^2) dx \\ &= x \left(c_1 + c_2 \left(-\frac{1}{x} - x \right) \right) = c_1 x - c_2 (1+x^2) \end{aligned}$$

【考点延伸】《考试宝典》专题六 4.4—欧拉方程

7. 【解析】令 $x = e^t$, 则 $\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt}$, $\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right) = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$

所以 $\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2y = \frac{1}{e^t + 1}$, 易知特征根为 $\lambda_1 = -1, \lambda_2 = -2$, $Y = c_1 e^{-t} + c_2 e^{-2t}$

利用常数变易法, 令 $y = c_1(t) e^{-t} + c_2(t) e^{-2t}$, $y' = c_1'(t) e^{-t} + c_2'(t) e^{-2t} - c_1(t) e^{-t} - 2c_2(t) e^{-2t}$

令 $c_1'(t) e^{-t} + c_2'(t) e^{-2t} = 0$, 则 $y'' = -c_1'(t) e^{-t} + c_1(t) e^{-t} - 2c_2'(t) e^{-2t} + 4c_2(t) e^{-2t}$

代入方程可得

$$\begin{aligned} &-c_1'(t) e^{-t} + c_1(t) e^{-t} - 2c_2'(t) e^{-2t} + 4c_2(t) e^{-2t} + 3(-c_1(t) e^{-t} - 2c_2(t) e^{-2t}) + 2(c_1(t) e^{-t} + c_2(t) e^{-2t}) \\ &= -c_1'(t) e^{-t} - 2c_2'(t) e^{-2t} = \frac{1}{1+e^t} \end{aligned}$$

$$\text{联立 } c_1'(t) e^{-t} + c_2'(t) e^{-2t} = 0 \text{ 可解得 } \begin{cases} c_1'(t) = \frac{e^t}{e^t + 1} \\ c_2'(t) = -\frac{e^{2t}}{e^t + 1} \end{cases} \Rightarrow \begin{cases} c_1(t) = \ln(e^t + 1) + d_1 \\ c_2(t) = \ln(e^t + 1) - e^t + d_2 \end{cases}$$

因此 $y = (\ln(e^t + 1) + d_1) e^{-t} + (\ln(e^t + 1) - e^t + d_2) e^{-2t}$

$$= \ln(e^t + 1) (e^{-t} + e^{-2t}) + (d_1 - 1) e^{-t} + d_2 e^{-2t}$$

$$= \ln(x+1) \left(\frac{1}{x} + \frac{1}{x^2} \right) + \frac{a_1}{x} + \frac{a_2}{x^2} \quad (a_1, a_2 \text{ 为常数})$$

【考点延伸】《考试宝典》专题六 题型 4—高阶线性微分方程的求解

8 【解析】当 $x=0$ 时, $f(0)=1$, 对原方程两边求导可得 $\int_0^x f(t) dt = -\sin x - f'(x)$, $f'(0)=0$

再次求导可得 $f(x) = -\cos x - f''(x)$, 即 $f''(x) + f(x) = -\cos x$

齐次方程得通解为 $F = c_1 \cos x + c_2 \sin x$, 设特解为 $f^*(x) = x(A \cos x + B \sin x)$, 代入原方程可得

$$A=0, B=-\frac{1}{2} \Rightarrow f(x) = c_1 \cos x + c_2 \sin x - \frac{1}{2} \sin x$$

$$f(0) = c_1 = 1, f'(0) = c_2 = 0 \Rightarrow f(x) = \cos x - \frac{1}{2} x \sin x$$

【考点延伸】《考试宝典》专题六 题型 5—微分方程的应用

9 【解析】令 $F(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}, A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ 方程即为 $\frac{dF}{dt} = AF$

$$\text{令 } D(\lambda) = \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = -\lambda^3 + 3\lambda + 2 = -(\lambda - 2)(\lambda + 1)^2 = 0, \text{得}$$

$\lambda = 2$ (单根), $\lambda = -1$ (二重根)

$$\text{当 } \lambda = 2, A - E\lambda = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}, \text{求得特征向量为 } p_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\text{当 } \lambda = -1, A - E\lambda = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \text{求得特征向量为 } p_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, p_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{因此 } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} e^{-t} + c_3 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{-t}$$

【考点延伸】《考试宝典》专题六 题型 5—微分方程的应用